RFM Customer Segmentation

& Customer Lifetime Value Analysis

A Data Analytics Case Study

Using Python • Power BI

Dataset: Online Retail II UCI | 1,067,371 transactions | 2009–2011

1. Background & Business Problem

Most ecommerce businesses collect a lot of transaction data, but not many actually use it to make smarter decisions about who to focus on. They send the same discount to every customer, spend the same amount trying to win back everyone, and treat a customer who bought once three years ago the same way they treat someone who orders every month.

This project is built around a very common but expensive problem in ecommerce:

> *"The company has thousands of customers but is allocating its retention budget equally across all of them, without knowing which customers are actually worth retaining."*

The goal of this project was to fix that. Using a method called RFM analysis, the idea was to look at how each customer actually behaved -- when they last bought, how often they bought, and how much they spent -- and use that behaviour to group them into meaningful segments. Then, for each segment, estimate how much that customer is worth over their lifetime, so the business knows where to put its money.

What questions this project answers

- Which customers are our best, and what makes them different from the rest?
- What percentage of customers are showing signs of going silent or already lost?
- How much is a Champion customer actually worth compared to a Lost one?
- Where should the retention budget go, and where should it not go?

2. About the Dataset

The dataset used is the Online Retail II UCI dataset, which is publicly available on Kaggle. It contains real transaction data from a UK-based online giftware retailer that primarily sells to wholesale buyers and individual customers across Europe.

Detail	Value
Total Rows	1,067,371 transactions
Columns	8
Time Period	December 2009 to December 2011 (2.02 years)
Geography	92% United Kingdom, 8% rest of Europe

Source	UCI Machine Learning Repository / Kaggle

The 8 Columns and How They Were Used

Column	Type	Role in This Project
Invoice	ID	Counting distinct purchases per customer (Frequency)
StockCode	ID	Product-level analysis
Description	Text	Understanding what products are being bought
Quantity	Number	Calculating revenue; negative = returns
InvoiceDate	Date/Time	Calculating Recency and monthly trends
Price	Number	Calculating revenue per transaction
Customer ID	ID	Core of all RFM calculations -- grouping by customer
Country	Text	Geographic segmentation

3. Tools Used

Tool	Purpose
Python (Pandas)	Data loading, cleaning, RFM calculation, CLV estimation
Matplotlib & Seaborn	Building all charts and visualizations
Jupyter Notebook	Development environment -- all code was written and run here
Power BI Desktop	Interactive 2-page dashboard

4. Phase 1 — Data Loading & Exploration

Before doing any analysis, the first step was just to understand the data. Load it, look at it, and figure out what problems exist before touching anything.

The dataset was loaded into Python using Pandas and immediately showed a few important things:

- Total rows: 1,067,371 across 8 columns
- Customer ID had only 824,364 non-null values out of 1,067,371 -- meaning 243,007 rows belonged to customers with no ID (guest checkouts)
- InvoiceDate was stored as text (object type), not as an actual date -- Python could not do date calculations on it yet
- Quantity had a minimum value of -80,995 -- returns and cancellations were mixed into the data
- Price also had negative values -- adjustment rows and errors existed in the dataset

These findings set up exactly what needed to be fixed in the cleaning phase.

5. Phase 2 — Data Cleaning

Raw data is almost never ready to analyse directly. The cleaning phase was about making a series of deliberate decisions about what to keep, what to remove, and what to handle separately -- and being able to explain the business reason for each decision.

What Was Cleaned and Why

Removing null Customer IDs. 243,007 rows had no Customer ID. RFM analysis requires grouping all transactions by customer -- without a Customer ID, a row cannot be linked to anyone. These rows represent guest checkouts and were dropped. They went from 1,067,371 rows down to 824,364.

Separating returns (negative quantity). 18,744 rows had negative Quantity values -- these are product returns and cancellations. Rather than deleting them, they were saved as a separate file (returns_data.csv). Returns data has its own analytical value -- which products get returned most, which customers return a lot -- and discarding it would have been wasteful.

Removing zero and negative prices. Only 71 rows had zero or negative Price values. These are typically internal test orders or accounting adjustments. They were removed because a product sold for £0 does not contribute to any revenue calculation.

Converting InvoiceDate from text to date format. Python was reading the InvoiceDate column as plain text. To calculate how many days ago a customer last purchased (which is

Recency), Python needs to understand the column as a date, not a string. This was converted using pd.to_datetime().

Creating a Revenue column. No revenue column existed in the raw data. Revenue was calculated as Quantity x Price for each row. This became the foundation of the Monetary dimension in RFM.

After Cleaning

Check	Result
Final rows	805,549
Columns	9 (original 8 + new Revenue column)
Null values	0 across all columns
Negative quantities	0 (moved to returns file)
Total Revenue in dataset	£17,743,429.18

6. Phase 3 — RFM Calculation

RFM stands for Recency, Frequency, and Monetary. It is one of the most widely used customer segmentation frameworks in ecommerce and retail because it is based purely on actual purchase behaviour, not assumptions or demographics.

What Each Dimension Measures

Dimension	What It Measures	How It Was Calculated
Recency (R)	How many days since the customer last bought	Days between last InvoiceDate and reference date (Dec 10, 2011)
Frequency (F)	How many separate orders the customer placed	Count of distinct Invoice numbers per customer
Monetary (M)	Total amount the customer spent	Sum of Revenue across all their transactions

Scoring Logic

Each customer received a score of 1 to 4 on each dimension using quartile-based scoring (pd.qcut). Customers were divided into 4 equal groups:

- For Recency: lower days = better = score of 4. A customer who bought 2 days ago gets R=4. A customer who bought 400 days ago gets R=1.
- For Frequency and Monetary: higher = better = score of 4.
- The three scores were added together to create a combined RFM Score ranging from 3 to 12.

Segment Labels

Based on the R, F, and M scores, each customer was labelled into one of 6 named segments using a rule-based classification:

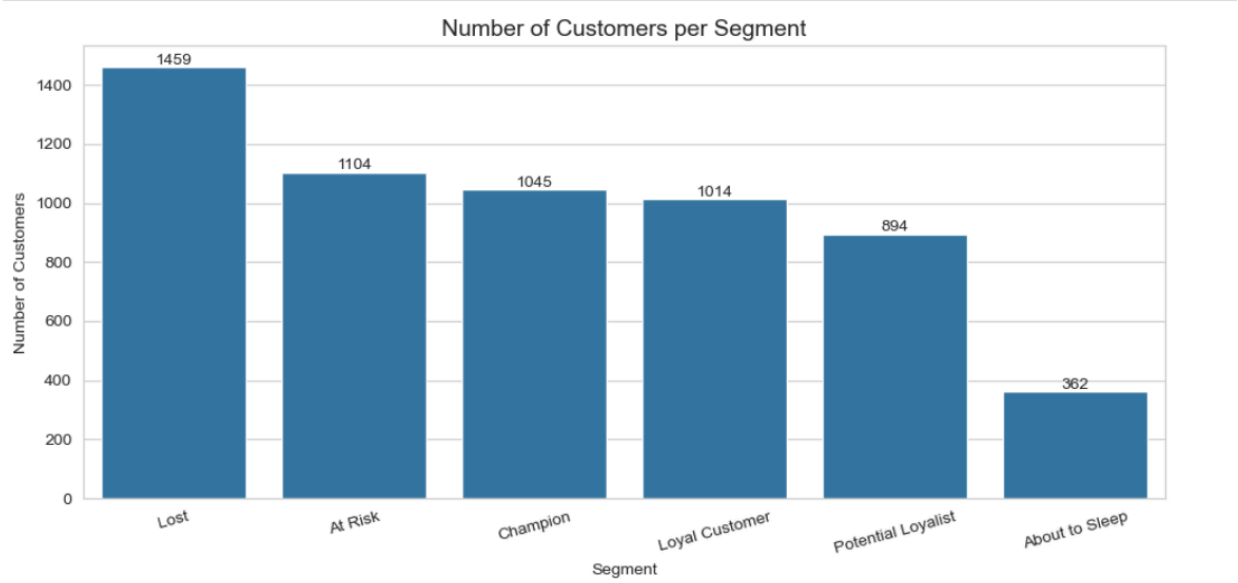
Segment	Score Logic	Business Meaning
Champion	R≥4, F≥3, M≥3	Bought recently, buys often, spends most
Loyal Customer	R≥3, F≥3	Regular buyers with consistent spend
Potential Loyalist	R≥3, F≤2	Recent buyers who haven't bought often yet
At Risk	R=2, F≥2	Used to buy often but going cold
About to Sleep	R=2, F=1	Low engagement, fading away
Lost	All remaining (R=1)	Haven't bought in a very long time

7. Phase 4 — Exploratory Data Analysis & Visualizations

With the RFM scores and segments calculated, the next step was to visualize the data and extract the key business insights. Each chart below was built to answer a specific business question.

Chart 1: Number of Customers per Segment

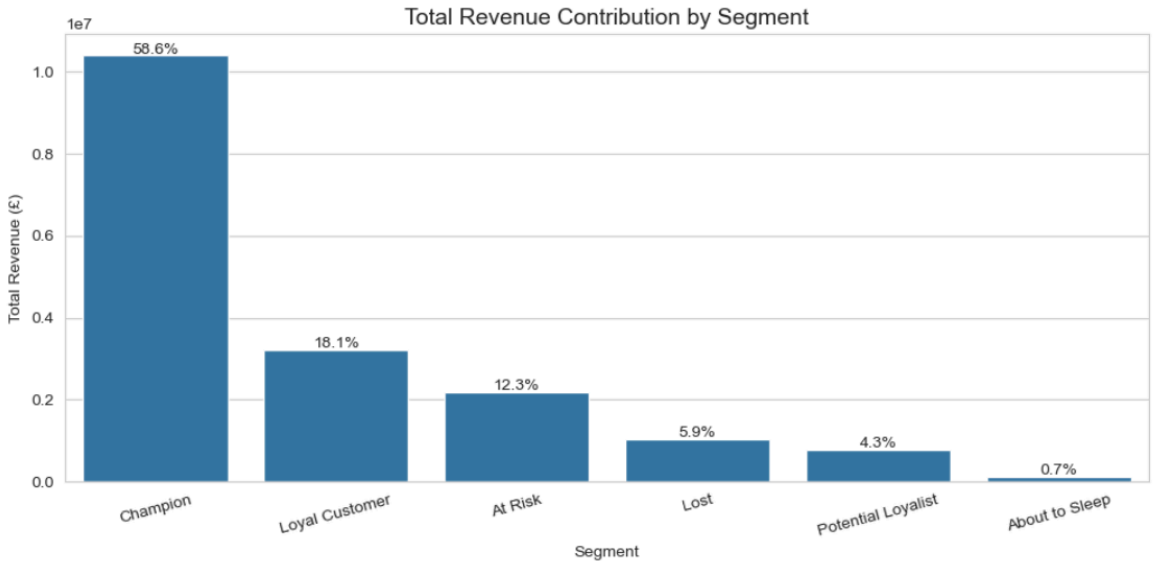
How are customers distributed across segments?



The largest segment is Lost at 1,459 customers (24.8%), followed closely by At Risk at 1,104 (18.8%). Champions are 1,045 customers (17.8%). The fact that Champions are the third largest segment -- despite being the most demanding to acquire -- suggests this retailer has done a reasonable job of building a loyal base. However, the combined churn signal from Lost and At Risk is significant: 43.6% of the customer base is either gone or going.

Chart 2: Revenue Contribution by Segment

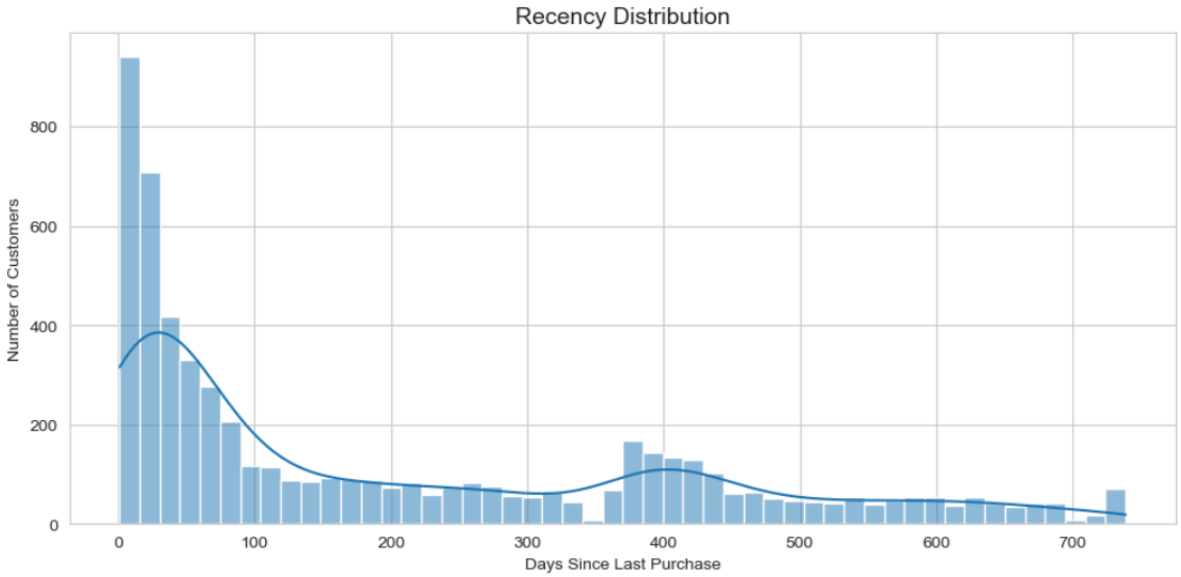
Which segments are actually generating the revenue?



This is the most important chart in the project. Champions represent 17.8% of the customer base but generate 58.6% of all revenue. Add Loyal Customers (18.1%) and together just 35% of customers are responsible for 76.7% of total revenue. The Lost segment -- the largest by customer count -- contributes only 5.9% of revenue, confirming that customer count alone is a misleading metric for business health.

Chart 3: Recency Distribution

When did customers last purchase?

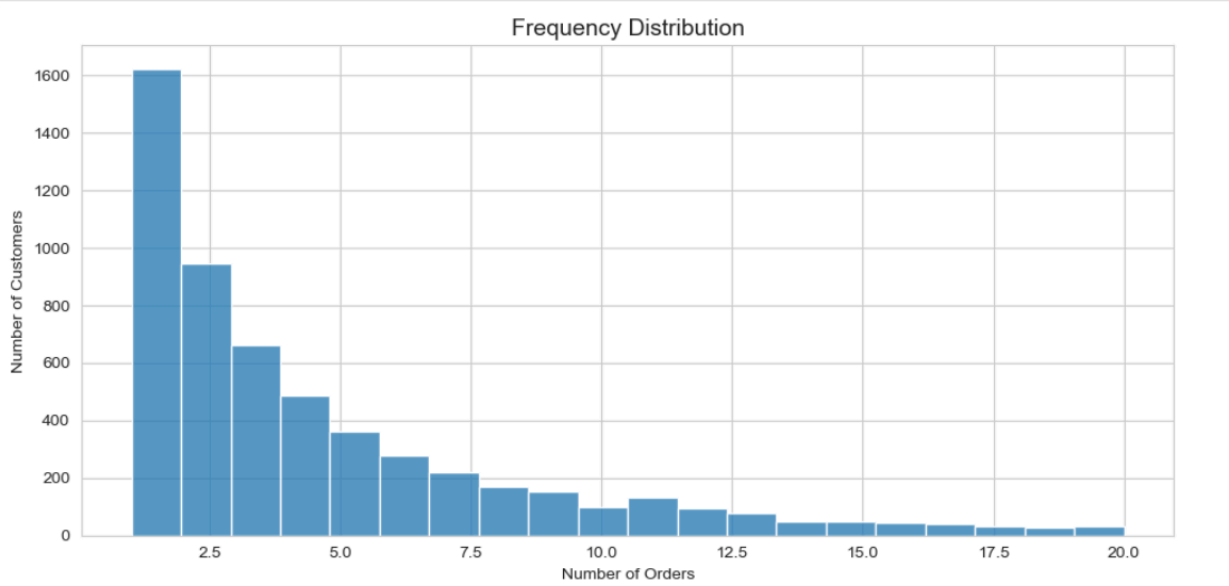


The chart shows two distinct clusters. The first is a large spike on the left -- customers who bought very recently (within 0-100 days). These are largely Champions and Loyal Customers

who are still actively engaging. The second bump around 350-420 days is the Lost cluster -- customers whose last purchase was roughly a year ago and who have not returned since. The gap between these two groups tells you that customer engagement is quite polarized: customers are either active or they have churned, with relatively few in a genuine middle ground.

Chart 4: Frequency Distribution

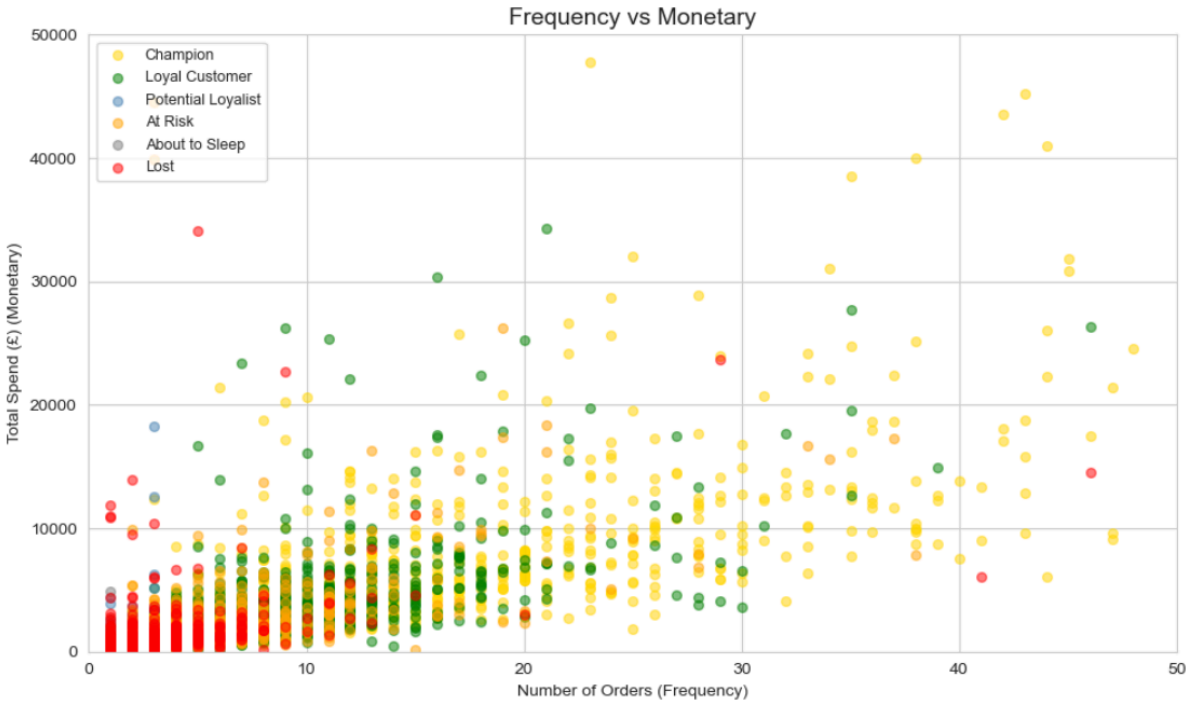
Are most customers one-time buyers or repeat buyers?



The most striking feature here is the massive bar at 1-2 orders. The majority of customers in this dataset have placed only one or two orders. This is extremely common in retail and represents the single biggest retention opportunity: converting a one-time buyer into a repeat buyer costs far less than acquiring a new customer. The long tail to the right shows a smaller group of highly frequent buyers -- these are predominantly Champions and Loyal Customers who have been buying consistently over the two-year period.

Chart 5: Frequency vs. Monetary (Who Are the High-Value Customers?)

Do customers who buy more also spend more, and who are the outliers?



Gold dots (Champions) dominate the upper right quadrant -- high frequency and high spend. They are clearly separable from the rest. Red dots (Lost) cluster in the lower left -- they bought rarely and spent little before disappearing. One notable outlier is a red Lost dot sitting around £34,000 spend at just 3 orders -- a high-value customer who made large purchases in a short window and then vanished. This is a real business problem: a whale who was not identified and retained in time. The scatter plot makes this kind of anomaly immediately visible in a way a table never would.

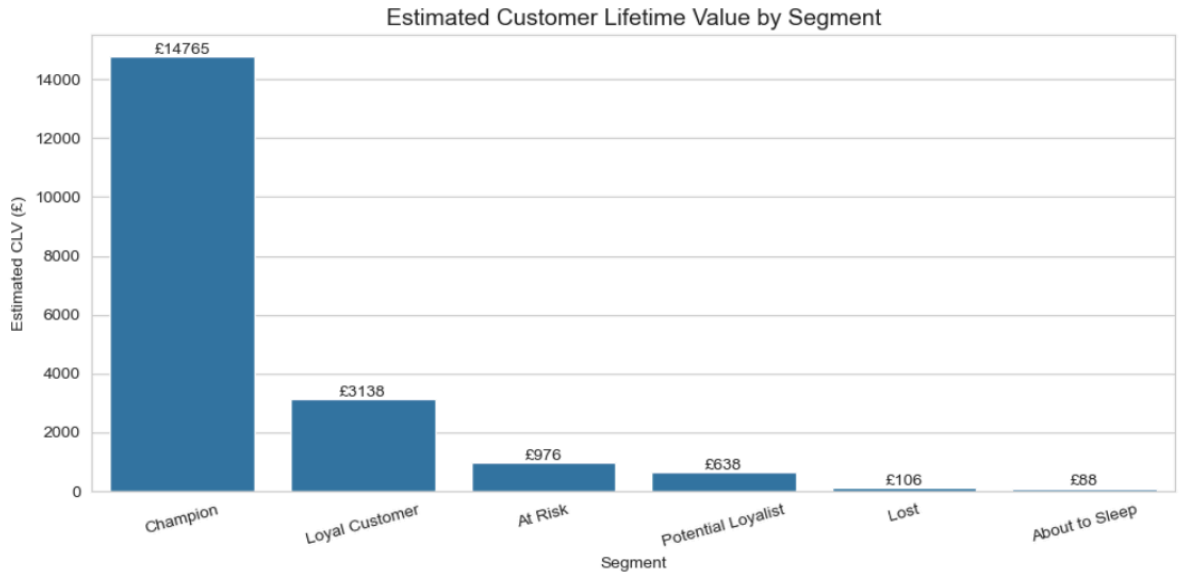
8. Phase 5 — Customer Lifetime Value (CLV) Estimation

RFM tells you who your customers are right now. CLV tells you how much they are worth over time. This is the number that marketing and finance teams use to decide how much budget to allocate to acquiring and retaining different customer groups.

The Formula

> **CLV = Average Order Value × Orders Per Year × Customer Lifespan (years)**

The three ingredients were calculated directly from the data for each segment. Customer lifespan was estimated based on segment behaviour patterns -- Champions were assigned 3 years (they are still active and loyal), while Lost customers were assigned 0.3 years (their relationship with the business is effectively over).



CLV Results by Segment

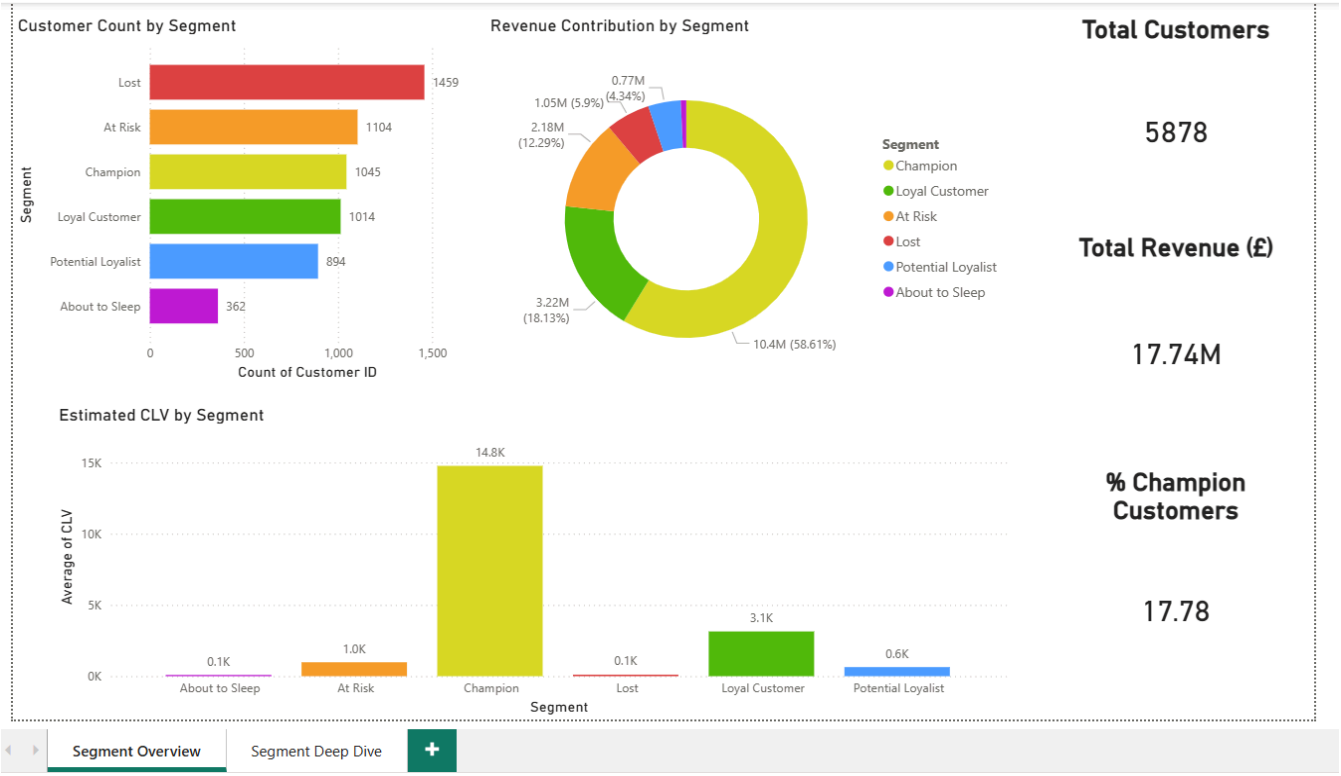
Segment	Avg Order Value	Orders/Year	Estimated CLV
Champion	£561.64	8.76	£14,765
Loyal Customer	£398.02	3.94	£3,138
At Risk	£397.51	2.46	£976
Potential Loyalist	£484.25	0.88	£638
Lost	£356.93	0.99	£106
About to Sleep	£358.40	0.49	£88

The gap between Champions (£14,765) and Lost customers (£106) is 139x. This single comparison is the clearest argument for why differentiated retention strategy matters -- and why spending equal budget on all customers is a waste of money.

9. Phase 6 — Power BI Dashboard

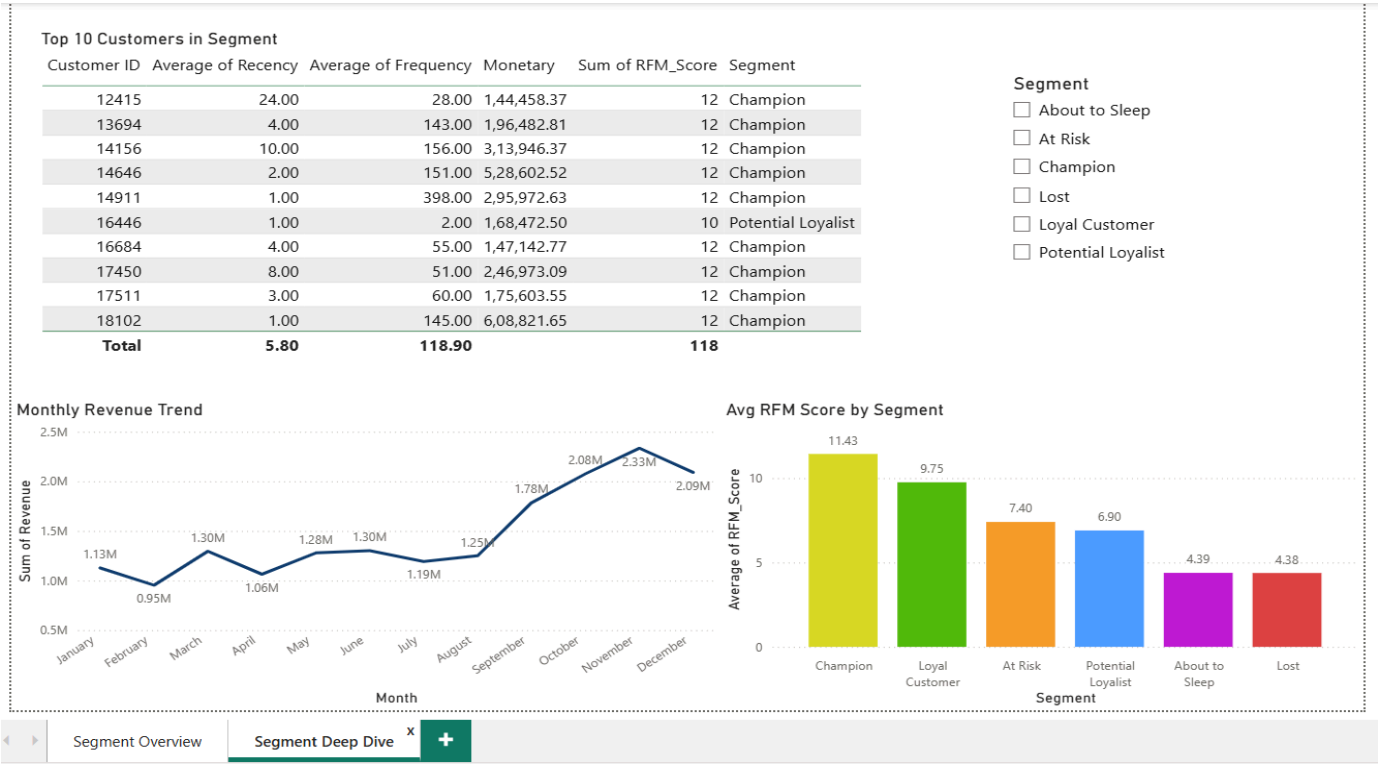
The final phase was building a two-page interactive dashboard in Power BI that brings all the analysis together. All three data files -- rfm_output.csv, clv_output.csv, and online_retail_cleaned.csv -- were loaded and connected through relationships on Customer ID and Segment.

Page 1 — Segment Overview



Page 1 gives a top-level picture of the customer base. The three KPI cards show total customers (5,878), total revenue (£17.74M), and the percentage of Champion customers (17.78%). The bar chart on the left shows customer counts by segment. The donut chart in the centre shows revenue contribution -- the yellow slice (Champions at 58.61%) visually dominates the chart. The CLV column chart at the bottom shows the steep drop in value from Champions to all other segments.

Page 2 — Segment Deep Dive



Page 2 is designed for exploring individual segments. The slicer on the right allows filtering by any segment, which dynamically updates all four visuals. The table shows the top 10 customers by monetary value in the selected segment. The monthly revenue trend line chart shows a clear seasonal pattern -- revenue is relatively flat from January through August, then climbs sharply from September, peaking in November at £2.33M before dipping slightly in December. This is the holiday season effect, and it is a meaningful operational insight for the business. The average RFM score bar chart confirms the segmentation logic is clean -- Champions average 11.43, descending logically all the way down to Lost at 4.38.

10. Key Findings

Finding	What It Means
17.8% of customers generate 58.6% of revenue	Classic Pareto -- the business is heavily dependent on a small group of Champions
43.6% of customers show churn signals	Lost + At Risk combined = 2,563 customers. This is the most urgent problem to address
Champion CLV is 139x that of a Lost customer	Equal retention spend across all segments is irrational given this gap
Most customers placed only 1-2 orders	Converting one-time buyers to repeat buyers is the biggest growth lever
Revenue peaks in November (£2.33M)	Strong seasonal dependency -- the business needs to ensure Champions and Loyals are nurtured before the holiday season each year
At Risk customers have a CLV of £976	Still valuable enough to justify a targeted win-back campaign before they slide into Lost

11. Business Recommendations

These recommendations are not generic. They are based directly on the segment behaviour patterns, revenue contribution percentages, and CLV figures that came out of this analysis.

Champions — 1,045 customers, 58.6% of revenue, CLV £14,765

These customers do not need to be converted -- they are already there. The risk with Champions is complacency: assuming they will keep buying without any deliberate effort. Given their CLV, it is worth treating them like VIPs. Early access to new products, a loyalty reward tier, and personalised communication (based on what they actually buy) would reinforce the relationship. Do not send Champions generic promotional emails -- they are past that stage and it feels tone-deaf.

Loyal Customers — 1,014 customers, 18.1% of revenue, CLV £3,138

Loyal Customers are buying regularly but have not crossed into Champion territory yet. The gap between their CLV (£3,138) and Champion CLV (£14,765) is largely explained by order

frequency -- Champions place 8.76 orders per year vs 3.94 for Loyals. The goal here is to increase purchase frequency, not order value. A trigger-based email (such as "it's been 3 weeks since your last order -- here's what's new") is more effective for this group than a blanket discount.

At Risk — 1,104 customers, 12.3% of revenue, CLV £976

At Risk customers used to buy with some regularity but their Recency score has dropped. A CLV of £976 means they still have meaningful value, and acting before they become Lost is significantly cheaper than trying to recover them afterward. A win-back campaign with a time-limited offer (not a permanent discount, which trains customers to wait for deals) is the right approach here. The intervention window is narrow -- once they hit the Lost category, the probability of recovery drops sharply.

Potential Loyalists — 894 customers, 4.3% of revenue, CLV £638

These are recent buyers who have not bought frequently yet. This is actually one of the most promising segments for long-term growth because the relationship is new and the customer is still forming habits. A structured post-purchase follow-up sequence -- a thank you email on day 3, a product recommendation on day 10, a review request on day 14 -- can meaningfully increase the probability that their second purchase happens, which is the most critical conversion in any retail customer lifecycle.

Lost — 1,459 customers, 5.9% of revenue, CLV £106

A CLV of £106 makes aggressive spend on this group hard to justify. A single low-cost re-engagement email (perhaps tied to a seasonal moment like Christmas or a product they bought before going back in stock) is reasonable. Beyond that, the marketing budget is better spent on At Risk customers who still have meaningful value left. The key insight here is not that Lost customers do not matter -- it is that the cost of recovery almost always exceeds the expected return at this CLV level.

About to Sleep — 362 customers, 0.7% of revenue, CLV £88

Similar to Lost. Minimal spend recommended. These customers have very low engagement and their contribution is negligible. A one-time automated email is the maximum worthwhile investment for this group.

12. Conclusion

This project started with a simple but expensive problem: a company spending its retention budget without knowing which customers were actually worth retaining. RFM analysis gave that problem a data-driven answer.

By segmenting 5,878 customers based on their actual purchasing behaviour and estimating the lifetime value of each segment, the analysis surfaces three things that matter to any ecommerce business:

- Where the real revenue is coming from (Champions, 17.8% of customers, 58.6% of revenue)
- Where the biggest retention risk lies (Lost + At Risk, 43.6% of customers and growing)
- How much a customer is worth over time, by segment, so budget decisions can be made rationally rather than by gut feel

The tools used- Python for calculation and Power BI for presentation. The analysis is reproducible, the dashboard is interactive, and the recommendations are tied directly to the numbers rather than being generic advice.

For any ecommerce business sitting on transaction data but not segmenting its customers, this type of analysis is one of the highest-return analytical investments available. The data is already there. The cost is time.